

Python: Library Catalog System

Create a system to manage different types of items in a library catalog, such as books and DVDs. Implement classes for each item type with specific attributes and methods to display information.

The system should utilize magic methods like `__str__` and `__repr__` for enhanced functionality.

Implement the following classes:

1. **Item Class** with attributes for title, author, and year. It should include methods to:
2. Return a formatted string representation using **str**: "[title] by [author] ([year])"
3. Return a formal representation using **repr**: "Item: [Title] by [Author] ([Year])"

Method: **display_info** to display detailed information about the item. `text Title: [title]
Author: [author] Year: [year]`

Book Class (inherits from Item) with additional attributes for genre and ISBN. It should override the **display_info** method to include book-specific information. `text Title: [title]
Author: [author] Year: [year] Genre: [genre] ISBN: [ISBN]`

DVD Class (inherits from Item) with an additional attribute for duration. It should override the **display_info** method to include DVD-specific information. `text Title: [title] Author: [author] Year: [year] Duration: [duration] minutes`

Constraints * The number of items is less than or equal to 50.

Input Format For Custom Testing The first line indicates the number of items. The next lines contain a comma-delimited list with the type of item and its attributes, one item per row.

Sample Case 0 Sample Input For Custom Testing

```
2
Book,Python Programming,John Doe,2022,Programming,978-3-16-148410-0
DVD,Movie,Jane Smith,2021,120
```

Sample Output

```
--- Item Information ---
Python Programming by John Doe (2022)
Title: Python Programming
Author: John Doe
```

```
Year: 2022
Genre: Programming
ISBN: 978-3-16-148410-0
```

```
Details:
Movie by Jane Smith (2021)
Title: Movie
Author: Jane Smith
Year: 2021
Duration: 120 minutes
```

Explanation There are two items. The first type of item is a book, with attributes of the Book class. The second type of item is a DVD, with attributes of the DVD class.

Sample Case 1 Sample Input For Custom Testing

```
1
DVD,Movie,John Doe,2019,120
```

Sample Output

```
--- Item Information ---
Movie by John Doe (2019)
Title: Movie
Author: John Doe
Year: 2019
Duration: 120 minutes
```

Explanation There is one item. It is a DVD with attributes of a DVD instance.